

Mohsen Mostafa

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Kaggle: kaggle.com/babydriver123 | ORCID: 0009-0004-4478-0317 (preprints)

Professional Summary

AI Team Lead & Research Engineer – 8+ years in AI/ML, Computer Vision, GenAI (LLMs, RAG, agents).
Proven track record of production deployments with measurable business impact.

Core Skills

Languages & Data: Python, Pandas, SQL

ML/DL: PyTorch, TensorFlow, XGBoost, scikit-learn

GenAI/CV: Hugging Face, Falcon, DeepSeek, Llama, RAG, ColBERT, YOLOv8, OpenCV, Mask2Former

Deployment: FastAPI, Docker, ONNX, TensorRT, PostgreSQL, Redis, Pinecone, GCP Vertex AI

Experience

AI Team Lead – GraviLog (Remote, Part-time) *Sep 2025 – Present*

- AI roadmap, CV/multimodal R&D, GPU inference optimization, mentoring.

Senior AI Developer – OMC Technologies (Remote) *May 2023 – Aug 2025*

- AI agents (Falcon, DeepSeek) → automated 3 workflows, manual effort ↓40%
- Multimodal healthcare AI → chatbot F1 ↑19%
- Latency 210ms → 145ms (↓30%) via streaming & FP16
- Fraud detection AUC ↑17%, false positives ↓22%
- FastAPI microservices: p99 <95ms @500 RPS
- Mentored 5 juniors → 2 promotions

AI Developer & Researcher – Nexgro (South Africa) *Apr 2020 – Apr 2023*

- Grain quality CV → 98% accuracy (6 types)
- Drone/satellite anomaly detection (NDVI+YOLOv8) → manual inspection ↓60%
- LLM RAG assistant on technical manuals

- Document automation → saved 15 person-hours/week
- RL logistics prototype → travel distance ↓12%

Data Scientist – Nexgro *Mar 2018 – Mar 2020*

- Forecast models → logistics costs ↓8%
- Predictive maintenance → prevented 3 breakdowns, saved ~\$45K
- Inventory optimization → spoilage ↓18%

Systems Administrator – Egyptian Co. *Oct 2010 – Mar 2018*

- IT infra for 200+ users → 99.8% uptime. Self-pivot to data science (2016-2018 certs).

Research Preprints (Selected)

- Geometric Deep Learning for Point Cloud Segmentation – *under revision for WACV 2027*
- Physics-Informed Neural Networks for Fluid Flow – *arXiv 2025*

Full list on ORCID.

Leadership & Mentorship

Mentored 5+ engineers (2 promotions); led cross-functional AI initiatives & internal LLM/MLOps workshops.

Education

Bachelor's in Information Systems – High Institute for Specialized Technological Studies, Egypt (2001–2005)

Certifications (Selected)

IBM Data Science, Deep Learning Specialization ([DeepLearning.AI](#)), TensorFlow Developer, TensorFlow Advanced Techniques (Google), TensorFlow Data and Deployment, Google Cloud MLOps, Total Data Quality (University Of Michigan), Research Methodology (University Of London) .

Languages

Arabic (native) | English (fluent C1/C2)